Mech Interp cheat sheet front · observe, intervene, decompose

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tested: TL 3.9.0 · sae-lens 6.54.12 · sklearn 1.9.1

Load a model, cache every activation NB1

```
import torch
from transformer_lens import HookedTransformer, utils
torch.set_grad_enabled(False)          # inference only
device = "cuda" if torch.cuda.is_available() else "cpu"
model = HookedTransformer.from_pretrained("gpt2-small",
                                          device=device)
prompt = "The Eiffel Tower is in the city of"
tokens = model.to_tokens(prompt)       # [1, pos], BOS first
model.to_str_tokens(prompt)  # ['<|endoftext|>', 'The', ...]
logits, cache = model.run_with_cache(tokens)
# logits: [batch, pos, d_vocab]
top = logits[0, -1].topk(5).indices
model.to_str_tokens(top)       # [' London', ' Paris', ...]
resid = cache["resid_post", 6]
# same tensor as cache["blocks.6.hook_resid_post"]
_, small = model.run_with_cache(   # cache only what you need
    tokens, names_filter="blocks.6.hook_resid_post")
```

Hook names `utils.get_act_name(name, L)`

Hook point	Shape	Meaning
`hook_embed`	[b, pos, d_model]	token embedding
`blocks.{L}.hook_resid_pre`	[b, pos, d_model]	stream into layer L
`blocks.{L}.hook_resid_mid`	[b, pos, d_model]	after attn, before MLP
`blocks.{L}.hook_resid_post`	[b, pos, d_model]	stream out of layer L
`blocks.{L}.attn.hook_pattern`	[b, head, q, k]	attention weights
`blocks.{L}.attn.hook_z`	[b, pos, head, d_head]	per-head output
`blocks.{L}.hook_attn_out`	[b, pos, d_model]	attn write to stream
`blocks.{L}.mlp.hook_post`	[b, pos, d_mlp]	MLP neurons
`blocks.{L}.hook_mlp_out`	[b, pos, d_model]	MLP write to stream
`ln_final.hook_normalized`	[b, pos, d_model]	input to unembed

Short names: `utils.get_act_name("z", 9)` → `blocks.9.attn.hook_z`; `cache["pattern", 9]`. GPT-2 small: 12 layers, 12 heads, d_model 768, d_head 64.

Logit lens: read any layer as a prediction NB1

```
for L in range(model.cfg.n_layers):
    r = cache["resid_post", L][:, -1:, :]   # last position
    lg = model.unembed(model.ln_final(r))   # [1, 1, d_vocab]
    tok = lg[0, -1].argmax().item()
    print(L, model.to_single_str_token(tok))
# all layers at once (final LayerNorm applied for you)
acc = cache.accumulated_resid(layer=-1, incl_mid=False,
                              apply_ln=True, pos_slice=-1)
lens_logits = acc @ model.W_U    # [n_layers+1, 1, d_vocab]
```

The lens is a *view*, not a cause: early layers were never trained to be read this way.

Sparse autoencoders with SAELens NB3

```
from sae_lens import SAE
# sae-lens v6 returns the SAE itself (no tuple)
sae = SAE.from_pretrained(release="gpt2-small-res-jb",
                          sae_id="blocks.8.hook_resid_pre",
                          device=device)
hook = sae.cfg.metadata.hook_name  # 'blocks.8.hook_resid_pre'
text = "The Golden Gate Bridge is in San Francisco"
_, cache = model.run_with_cache(text)
x = cache[hook]                    # [1, pos, d_model]
f = sae.encode(x)                  # [1, pos, d_sae], sparse
x_hat = sae.decode(f)              # [1, pos, d_model]
vals, idx = f[0, -1].topk(5)       # top latents, last token
rel_err = ((x_hat - x).norm() / x.norm()).item()
sae.W_dec.shape                    # [d_sae, d_model]
```

Gemma Scope 2 in NB3 (Gemma 3 1B IT, float32 on T4): `SAE.from_pretrained("gemma-scope-2-1b-it-res", "layer_17_width_16k_l0_medium")`, layer 17 of 26. Without a GPU NB3 falls back to GPT-2 + `gpt2-small-res-jb` automatically.

Activation patching: clean → corrupted NB2

```
from functools import partial
ctx = "When John and Mary went to the store,"
clean = ctx + " John gave a drink to"
corrupt = ctx + " Mary gave a drink to"
c_tok = model.to_tokens(clean)
x_tok = model.to_tokens(corrupt)       # must be same length
ans = model.to_single_token(" Mary")
wrong = model.to_single_token(" John")

def logit_diff(logits):        # > 0: model prefers " Mary"
    return (logits[0, -1, ans] - logits[0, -1, wrong]).item()

clean_logits, clean_cache = model.run_with_cache(c_tok)
CLEAN = logit_diff(clean_logits)
CORR = logit_diff(model(x_tok))

def patch_resid(act, hook, pos):  # hook(act, hook) -> act
    act[:, pos, :] = clean_cache[hook.name][:, pos, :]
    return act

name = utils.get_act_name("resid_pre", 8)
patched = model.run_with_hooks(
    x_tok, fwd_hooks=[(name, partial(patch_resid, pos=-1))])
recovered = (logit_diff(patched) - CORR) / (CLEAN - CORR)
# 0 = no effect, 1 = clean behaviour fully restored

def patch_head(z, hook, head):     # one head, via hook_z
    z[:, :, head, :] = clean_cache[hook.name][:, :, head, :]
    return z

z9 = utils.get_act_name("z", 9)
model.run_with_hooks(
    x_tok, fwd_hooks=[(z9, partial(patch_head, head=9))])

# persistent hook: stays until you remove it
model.add_hook(name, partial(patch_resid, pos=-1))
out = model(x_tok)
model.reset_hooks()
# scoped hook: removed automatically on exit
with model.hooks(fwd_hooks=[(name,
                             partial(patch_resid, pos=-1))]):
    out = model(x_tok)
```

```
# every (layer, position) of resid_pre in one call (slow on CPU)
from transformer_lens import patching
metric = lambda lg: lg[0, -1, ans] - lg[0, -1, wrong]
grid = patching.get_act_patch_resid_pre(
    model, x_tok, clean_cache, metric)    # [n_layers, pos]
```

Pitfalls. Clean and corrupted prompts must tokenize to the same length. A hook that edits in place must still `return` the tensor. A hook left by `add_hook` silently changes every later run: call `model.reset_hooks()`.

Steering: add a latent's decoder direction NB3

```
i, alpha = int(idx[0]), 8.0      # latent index, strength
v = sae.W_dec[i]                 # [d_model]
def steer(act, hook):
    return act + alpha * v       # all positions
with model.hooks(fwd_hooks=[(hook, steer)]):
    print(model.generate("I think that", max_new_tokens=20,
                         do_sample=False, verbose=False))
```

Sweep `alpha` (0, 2, 4, 8, 16) and compare with a *random* direction of the same norm: steering vectors are not unique.

Turn over: probes, calibration, circuit tracing, glossary, links

Mech Interp cheat sheet back · probe, audit, trace

Snippets continue from the front (same `model`, `utils`)
circuit-tracer 0.5.0: calls checked against the released source

Linear probe on the residual stream NB3 · NB5

```python
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
pos = ["I love this", "What a great day", "Amazing work",
       "Best meal ever", "I am so happy", "Truly delightful"]
neg = ["I hate this", "What an awful day", "Horrible work",
       "Worst meal ever", "I am so sad", "Truly disgusting"]
texts = pos + neg
y = np.array([1] * len(pos) + [0] * len(neg))
L = 6
hp = utils.get_act_name("resid_post", L)
def feats(t):                  # last-token residual at layer L
    _, c = model.run_with_cache(t, names_filter=hp)
    return c[hp][0, -1].float().cpu().numpy()
X = np.stack([feats(t) for t in texts])     # [n, d_model]
X_tr, X_te, y_tr, y_te = train_test_split(
    X, y, test_size=0.33, stratify=y, random_state=0)
probe = LogisticRegression(max_iter=1000, C=1.0)
probe.fit(X_tr, y_tr)
p = probe.predict_proba(X_te)[:, 1]   # P(class 1)
acc = probe.score(X_te, y_te)
```

Compare with a baseline (shuffled labels, layer 0, bag of words). When transfer matters, split by *rule or source*, not by row.

Calibration and selective prediction NB5

```python
from sklearn.metrics import roc_auc_score, brier_score_loss
from sklearn.calibration import calibration_curve
# conf: P(verdict correct); correct: 1 if the judge was right
rng = np.random.default_rng(0)
conf = rng.uniform(0.5, 1, 200)
correct = (rng.uniform(size=200) < conf).astype(int)

auroc = roc_auc_score(correct, conf)     # 0.5 = chance
brier = brier_score_loss(correct, conf)  # lower is better

def ece(conf, correct, n_bins=10):  # expected calib. error
    ids = np.clip((conf * n_bins).astype(int), 0, n_bins - 1)
    gap = 0.0
    for b in range(n_bins):
        m = ids == b
        if m.any():
            d = abs(correct[m].mean() - conf[m].mean())
            gap += d * m.mean()
    return gap

# reliability diagram: observed accuracy per confidence bin
frac_ok, mean_conf = calibration_curve(correct, conf, n_bins=10)

order = np.argsort(-conf)                # risk-coverage
n = np.arange(1, len(conf) + 1)
coverage = n / len(conf)
risk = np.cumsum(1 - correct[order]) / n

t = 0.8                          # threshold policy
auto = conf >= t                 # auto-accept; rest -> human
esc = (1 - correct[auto]).mean()  # error among accepted
print(f"coverage {auto.mean():.2f}, error {esc:.2f}")
```

Choose `t` on a held-out calibration split, never on the test set. Report coverage and error among accepted together.

Circuit tracing with circuit-tracer NB4

```python
# fresh runtime; NB4 pins (nnsight 0.8 breaks 0.5.0):
# %pip install circuit-tracer==0.5.0 nnsight==0.7.0
#   transformers==4.57.3 huggingface_hub==0.36.2
#   transformer_lens==3.2.1
from circuit_tracer import ReplacementModel, attribute, Graph
from circuit_tracer.utils import create_graph_files
# "gemma" = Gemma Scope transcoders for Gemma 2 2B (gated)
rm = ReplacementModel.from_pretrained(
    "google/gemma-2-2b", "gemma", dtype=torch.bfloat16)
# Gemma 3 (nnsight backend, experimental), e.g.
# ReplacementModel.from_pretrained("google/gemma-3-1b-it",
#   "mwhanna/gemma-scope-2-1b-it/transcoder_all/"
#   "width_16k_l0_small", backend="nnsight")
prompt = "Fact: the capital of the state containing Dallas is"
graph = attribute(prompt, rm, max_n_logits=10,
                  desired_logit_prob=0.95, batch_size=256,
                  max_feature_nodes=4096, verbose=True)
graph.to_pt("dallas.pt")       # Graph.from_pt("dallas.pt")
create_graph_files(graph, slug="dallas",
                   output_path="graphs/",   # pruned JSON
                   node_threshold=0.8, edge_threshold=0.98)
# validate: (layer, position, feature_idx, new_value)
logits, acts = rm.feature_intervention(
    prompt, [(20, 10, 15589, 0.0)])
```

NB4 default `PATH = "neuronpedia"`: no download. The Gemma 2 2B path is opt-in (~18 GB). Featured Dallas graph: see Links. `attribute(prompt, model, ...)`: prompt first. Rows of `graph.active_features` are (layer, pos, feature). Big *error nodes* = computation the transcoders miss. Ids above show the format only.

Glossary

Residual stream	The running sum every layer reads from and writes to; per-token shared memory.
Logit lens	Final LayerNorm + unembedding applied to an intermediate residual: its current guess.
Activation patching	Copy a clean-run activation into a corrupted run; measure how much behaviour returns.
Logit difference	logit(correct) − logit(alternative); a clean metric for patching.
Polysemantic	One neuron responds to several unrelated concepts.
Superposition	More features than dimensions, stored as almost-orthogonal directions.
SAE / latent	Sparse autoencoder: activation = sparse sum of decoder directions (latents).
Transcoder	Like an SAE, but reads an MLP's input and predicts its output.
Attribution graph	Direct linear effects among tokens, transcoder features, error nodes and logits for one prompt.
Error node	What the transcoders fail to reconstruct at a layer and position: unexplained computation.
Probe	A small, usually linear, classifier trained on activations to read out a property.
Steering vector	A direction added to the residual stream to push behaviour; not unique.
Calibration	Stated confidence matches observed accuracy (80% sure → right 80% of the time).
Risk–coverage	Error among accepted items versus share accepted, as the threshold moves.
Escape rate	Wrong verdicts that pass the threshold and skip human review.

Links

Neuronpedia	`neuronpedia.org`
Dallas graph (NB4)	`neuronpedia.org/gemma-2-2b/graph?slug=gemma-fact-dallas-austin`
Transformer Explainer	`poloclub.github.io/transformer-explainer`
ARENA, chapter 1	`learn.arena.education`
TransformerLens	`transformerlensorg.github.io/TransformerLens`
SAELens	`decoderesearch.github.io/SAELens`
circuit-tracer	`github.com/decoderesearch/circuit-tracer`
Gemma Scope 2	`huggingface.co/collections/google/gemma-scope-2`
Open problems	`arXiv 2501.16496 (Sharkey et al.)`

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